

R • TA

BDA400 • DATA SCIENCE TOOLS AND TECHNIQUES

TECHNICAL ANALYSIS USING R

Assignment 5 | Development Stage

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ASSIGNMENT	Development Stage • 15%
DATA SET	AAPL • MSFT • GOOGL
REPOSITORY	Open the Assignment 5 code on GitHub

Base R implementation • reproducible calculations • documented run steps

Prepared for academic review

Implementation Overview

The project implements the required technical indicators in separate R scripts, then applies them to the daily stock data from Assignment 2. Indicator calculations use base R; no technical-analysis package is required.

Indicator functions

SCRIPT / FUNCTION	WHAT IT DOES	OUTPUT
01_sma.R | sma()	Rolling arithmetic mean over each complete window.	Numeric vector; complete windows only.
02_ema.R | ema()	Recursive exponential average, seeded with the first value.	Numeric vector aligned to input.
03_macd.R | macd()	Short EMA minus long EMA; computes signal and histogram.	List: MACD, signal, histogram.
04_stdev.R | stdev()	Population standard deviation, dividing by n.	One numeric value.
05_linreg.R | linreg()	Least-squares line for a selected trailing window.	List: slope, intercept, predictions.
06_rsi.R | rsi()	Relative Strength Index using Wilder smoothing.	Aligned numeric vector; leading NA.
07_stoch_rsi.R | stoch_rsi()	Normalizes RSI to rolling extrema; smooths %K and %D.	List: aligned %K and %D.
08_crossover.R | crossover()	Detects upward/downward relative-position changes.	Up, Down, or None.
09_crossunder.R | crossunder()	Detects when first series crosses below second.	True, False, or None.

Project structure

functions/ contains the nine indicator scripts and the filters/trading-rules script. 00_run_assignment5.R loads the functions and runs the analysis. 99_validate_functions.R checks expected examples. The generated CSV files are kept in outputs/.

The Assignment 2 folder remains unchanged. Assignment 5 reads its OHLCV data and writes its own results into Assignment5/outputs.

Filters and Trading Rule

The extra workflow demonstrates how the indicators can be used to screen recent observations and create a simple position signal.

Example stock filter

For each ticker, the runner selects observations from the latest 90 calendar days and keeps rows where the closing price is above the independently calculated base-R SMA20. The filter function also supports optional date boundaries and a minimum-volume condition.

Moving-average trading rule

The position rule compares SMA20 with SMA50. An upward crossover enters a Long position; a downward crossover returns the position to Flat. The event and current position are included in the stock-level CSV output.

No market data is downloaded. The runner reads the daily OHLCV CSV files already stored in ../Assignment2/outputs for AAPL, MSFT, and GOOGL.

Run the project in RStudio

01	Set the working directory Open the repository in RStudio and set the working directory to the Assignment5 folder.
02	Validate the functions Run 99_validate_functions.R using Source, or enter source("99_validate_functions.R") in the Console. The expected message is: “All Assignment 5 indicator checks passed.”
03	Run the analysis Run 00_run_assignment5.R using Source, or enter source("00_run_assignment5.R").
04	Review the results Check the summary in the Console and the seven generated CSV files in Assignment5/outputs/.

Calculation Notes and Submission

The following conventions document how the implementation handles alignment and edge cases.

Calculation conventions

SMA

Returns values for complete windows only. The runner pads the earlier dates with NA so indicator values stay aligned with the trading dates.

EMA and MACD

EMA follows the assignment’s recursive formula and uses the first observation as its seed. MACD uses 12-, 26-, and 9-period EMAs.

Standard deviation

stdev() uses population standard deviation (division by n), as specified in the assignment pseudocode.

RSI

Uses Wilder smoothing and remains NA until the first complete period. If average gain and loss are both zero, RSI is 50; if losses alone are zero, it is 100; if gains alone are zero, it is 0.

StochRSI

Normalizes RSI using rolling highs and lows over the selected period, then smooths %K and %D. A flat range produces a raw value of 0.

Cross signals

crossover() returns Up, Down, or None. crossunder() returns True, False, or None, matching the supplied pseudocode.

Submission checklist

1. Upload this PDF to the LMS using the requested filename: Aljarad_Mohammad_CA_BDA400_A05.pdf.

2. Use the clickable GitHub link on the cover page to review the separate Assignment5 folder, R scripts, validation script, README, and output files.

3. The repository includes the source code and generated results; the cover-page link opens the project repository.